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(1) Research Program/Project/Study Title:		Rice – Sweet Potato Puto
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	Sigma
Province:	Capiz
Region:	VI

(5) Classification Research: Development: ☐ Basic ☒ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
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(7) Sector/Commodity Agriculture/crops	(8) Discipline/Field of Specialization Food
(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.) Abstract Convenient health foods that impart extra value in the form of nutritious and functional foods are now the highest priority for local product development in the modern food industry. Rice is naturally gluten-free and contains proteins that are known to be nutritious and hypoallergenic. Sweet potato is an economical and healthy food crop that has been widely grown in the Philippines. It is also gluten-free and particularly rich in beta-carotene, potassium, dietary fiber, carbohydrates, and other nutrients. The objectives of this study was determine the sensory qualities of rice - sweet potato puto in terms of appearance, aroma, taste, texture; determine the general acceptability of the product and determine if there are significant differences in the sensory qualities and general acceptability of the products among treatments. This study was conducted at Capiz State University Sigma Satellite College, Sigma, Capiz during the School Year 2022-2023. This experimental study utilized CRD as research design with five Treatments replicated three times evaluated by 15 evaluators using nine-point Hedonic scale. The statistical tools used were mean and ANOVA. Results of the study revealed that Treatment C (50% sweet potato flour and 50% rice flour) got the highest mean score in terms of appearance, aroma, taste, and texture. It was also revealed that product in Treatment C was the most acceptable. Further revealed that there is highly significant differences among treatments in terms of its sensory qualities in favor of Treatment C. Lastly, it was also revealed that there is highly significant difference among Treatments in terms of its general acceptability in favor of Treatment C. It can be implied that the formulation 50% sweet potato flour and 50% rice flour in making puto makes the product more attractive, pleasant, liked very much, and moderately soft. It has a potential for commercialization. (9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.) Sensory qualities, general acceptability, nine-point Hedonic scale, analysis of variance	
(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews) Introduction	

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Convenient health foods that impart extra value in the form of nutritious and functional foods are now the highest priority for local product development in the modern food industry. Rice is naturally gluten-free and contains proteins that are known to be nutritious and hypoallergenic. Sweet potato is an economical and healthy food crop that has been widely grown in the Philippines. It is also gluten-free and particularly rich in beta-carotene, potassium, dietary fiber, carbohydrates, and other nutrients (Shih et al., 2005).

Rice cake is a relatively new snack. It is made in many cultures and have a wide range of processing and product characteristics. The Filipino rice cake, puto is consumed daily as a breakfast, dessert or snackfood. The product is made from rice that is soaked overnight, ground and mixed with sugar and coconut milk (Arendt and Zannini, 2013).

Starchy roots and tuber crops play a pivotal role in the human diet. There are number of roots and tubers which make an extensive biodiversity even within the same geographical location. Thus, they add variety to the diet in addition to offering numerous desirable nutritional and health benefits such as antioxidative, hypoglycemic, hypocholesterolemic, antimicrobial, and immunomodulatory activities (Vagilad et al., 2020).

In Asian countries, some edible tubers are also used as traditional medicinal. A variety of foods can be prepared using tubers and they may also be used in industrial applications. Tubers have an immense potential as functional foods and nutraceutical ingredients to be explored in disease risk reduction and wellness (Vagilad et al., 2020).

Sweet potato (*Ipomoea batatas* L.) is an important alternative source of carbohydrates and attains fourth place after rice, corn and cassava. Presently, this crop is considered as having low economic value but it has significant social importance. It is most versatile for snack food, but it is used as staple food or as a rice substitute in many countries. It has a large potential to be used as food in developing nations with limited resources because of short maturity time, ability to grow under diverse climatic conditions and on less fertile soil. Sweet potato flour can serve as a source of energy and carbohydrates, beta carotene (pro-vitamin A), minerals (Ca, P, Fe and K) and dietary fiber which can add natural sweetness, colors and flavor to processed food products (Saeed et al., 2012). Sweet potato (*Ipomoea batatas* L), can substitute for rice because they have carbohydrates that are not inferior to rice (Wijaya et al., 2021).

The objectives of this study was utilized sweet potato flour and rice flour in making puto. Specifically to determine the sensory qualities of rice-sweet potato puto in terms of appearance, aroma, taste, texture; determine the general acceptability of the product in terms of appearance, aroma, taste, texture; and determine if there are significant differences in the sensory qualities and general acceptability of the products among treatments.

(11) Materials and Methods or Methodology (This includes the materials, methods,duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

Methodology

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This study utilized experimental method of research. This was conducted at Capiz State University Sigma Satellite College during the school year 2022 – 2023. The materials and tools used in this study were the following: Thirty (30) rice cake/puto molder (standard size), one (1) set of measuring cup/spoon, one (1) unit spiral mixer, one (1) unit steamer, five (5) mixing bowl, utility plates, and weighing scale.

The procedure in extracting flour from sweet potato were the following: washing the potato, peeling, slicing into thin sizes, sun-drying, blending, and straining. The sweet potato was spread out in a clean, dry and flat surface to ensure even distribution of the heat under the sun. The sweet potato was dried in a direct exposure of sunlight for about two to three (2-3) days until they were easy to grind with the use of blender or rice milling machine. Straining the sweet potato flour using strainer in order to separate the big particles of sweet potato to powdered one. Then, transferring it to a clean container and now ready for use.

Procedure in making the rice-sweet potato puto. The rice flour and sweet potato flour were sifted and were mixed in a medium sized bowl. Add and mix the dry ingredients such as brown sugar, baking powder, and salt. Add gradually the evaporated milk, water, egg, melted butter and mix until combined, do not overmix. Pour into greased puto molds or small silicone muffin molds until 3/4 full and place in a prepared steamer, then steam for 20 minutes or until cooked. Remove from steamer then let it slightly cool, remove from molds, and serve/package.

The experimental design used in the study was the single factor experiment with Completely Randomized Design (CRD) in which rice-sweet potato puto was studied using five treatments, Treatment A - 100 % sweet potato flour, Treatment B – 75 % sweet potato flour and 25 % rice flour, Treatment C – 50 % sweet potato flour and 50 % rice flour, Treatment D – 25 % sweet potato flour and 75 % rice flour and Treatment E – 100 % rice flour. The scorecard using nine-point Hedonic scale was used to evaluate the sensory qualities such as appearance, aroma, taste, and texture; and general acceptability of the product. The fifteen evaluators were composed to five (5) teachers teaching food related subject, five (5) students, and five (5) consumers. Mean and ANOVA were the statistical tools used in analyzing the data.

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Results and Discussions

Table 1.0 Sensory qualities of rice-sweet potato puto.

Treatment	Mean	Verbal Interpretation
Appearance		
Treatment A	6.93	Moderately Attractive

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Treatment B	6.60	Moderately Attractive
Treatment C	8.33	Extremely Attractive
Treatment D	6.60	Moderately Attractive
Treatment E	6.80	Moderately Attractive

Aroma

Treatment A	6.67	Moderately Pleasant
Treatment B	6.60	Moderately Pleasant
Treatment C	8.07	Very Much Pleasant
Treatment D	6.67	Moderately Pleasant
Treatment E	6.40	Moderately Pleasant

Taste

Treatment A	7.13	Liked Moderately
Treatment B	7.47	Liked Very Much
Treatment C	8.40	Liked Very Much
Treatment D	7.07	Liked Moderately
Treatment E	6.60	Liked Moderately

Texture

Treatment A	6.80	Moderately Soft
Treatment B	7.00	Moderately Soft
Treatment C	8.40	Moderately Soft
Treatment D	7.40	Moderately Soft
Treatment E	6.27	Slightly soft

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Sensory Qualities of Rice-Sweet Potato Puto

Table 1.0 deals with the sensory qualities of rice-sweet potato puto in terms of appearance, aroma, taste, and texture. In terms of appearance, product in Treatment C (50% sweet potato flour and 50% rice flour) was found “Extremely Attractive” with the mean score of 8.33, while, product in Treatment A (100% sweet potato flour) M=6.93; Treatment E (100% rice flour) M=6.80; Treatment B (75% sweet potato flour 25% rice flour) M= 6.60; and Treatment D (25% sweet potato flour and 75% rice flour) M= 6.60 were found “Moderately Attractive”. It implies that among the Treatments evaluated, products in Treatment C was extremely attractive as rated by the evaluators.

Same Table shows that in terms of aroma, Treatment C was found “Very Much Pleasant” with the mean score of 8.06, while Treatment A (M= 6.66), Treatment D (M= 6.66), and Treatment B (M=6.60), were found “Moderately Pleasant”, and Treatment E (M=6.40), “Slightly Pleasant”. It implies that product in Treatment C got the highest rating and was rated as very much pleasant. This result was in consonance to the findings of Vagilidad et al. (2020) revealed that the aroma of the sweet potato puto was highly acceptable, and there is no aftertaste of bitterness.

With regards to taste, Table shows that Treatment C was found “Liked Extremely” with the mean score of 8.40, while Treatment B (M=7.46) “Liked Very Much”; while, Treatment A (M= 7.13), Treatment D (M= 7.06), and Treatment A (M=6.60), were found “Liked Moderately” by the evaluators. It implies that among the treatments evaluated, product in Treatment A got the highest mean score and was liked very much by the panelist.

According to Vagilidad et al. (2020) in their study, in terms of taste, the respondents rated the sweet potato puto as highly acceptable.

Same Table shows that in terms of texture, Treatment C was found “Very Much Soft” with the mean score of 8.40, while Treatment D (M=7.40), Treatment B (M=7.00) and Treatment A (M=6.80), were found “Moderately Soft”, and, Treatment E with the mean score of 6.26 was found “Slightly Soft”. It implies that Treatment C got the highest mean score among other Treatments. Findings of the study of Shih et al. (2005) revealed that rice-sweet potato pancakes appeared to have the best combination of textural properties when sweet potato was added in the range of 20 – 40 %.

Table 2.0. General acceptability of rice-sweet potato puto.

Treatment	Appearance	Aroma	Taste	Texture	Grand Mean	Verbal Interpretation
Treatment A	6.93	6.67	7.13	6.80	6.88	Moderately Acceptable

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Treatment B	6.60	6.60	7.47	7.00	6.92	Moderately Acceptable
Treatment C	8.33	8.07	8.40	8.40	8.30	Very Much Acceptable
Treatment D	6.60	6.67	6.60	6.27	6.53	Moderately Acceptable
Treatment E	6.80	6.40	6.60	6.27	6.52	Moderately Acceptable

General Acceptability of Rice-Sweet Potato Puto

Data in Table 2.0 revealed the general acceptability of rice-sweet potato puto. Results revealed that rice-sweet potato puto made from Treatment C got the highest mean of 8.30 verbally interpreted as “Very Much Acceptable” while, product in Treatment B (M=6.92); Treatment A (M=6.88); Treatment D (M=6.53); and Treatment E (M=6.52) were moderately acceptable as rated by the evaluators. This indicated that Treatment C was the most acceptable product by the evaluators as indicated by its mean. It implies that the rice-sweet potato puto with the formulation of 50% sweet potato flour and 50% rice flour was highly accepted. This result agrees to the findings of the study of Vagilidad et al. (2020) who revealed that the level of acceptability of sweet potato puto was highly acceptable.

Table 3.0 Analysis of variance of the sensory qualities and general acceptability of rice-sweet potato puto

Source of Variance	Sum of Squares	df	Mean Square	F	Sig.
Appearance					
Between Groups	31.920	4	7.980	5.484*	.001
Within Groups	101.867	70	1.455		
Total	133.787	74			
Aroma					
Between Groups	27.120	4	6.780	4.708*	.002
Within Groups	100.800	70	1.440		
Total	127.920	74			
Texture					
Between Groups	38.213	4	9.553	6.050*	.001
Within Groups	110.533	70	1.579		
Total		74			

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Taste	27.067	4	6.767	7.221*	.001
Between Groups	65.600	70	.937		
Within Groups	92.667	74			
Total					

General					
Acceptability	28.322	4	7.080	7.240*	.001
Between Groups	68.458	70	.978		
Within Groups	96.780	74			
Total					

Differences on Sensory Qualities and General Acceptability of Rice-Sweet Potato Puto

Table 3.0 presents the Analysis of Variance on the sensory qualities such as appearance, aroma, taste, and texture and general acceptability of rice-sweet potato puto.

In terms of appearance, data show that there is a highly significant difference among Treatments ($F=5.484$; $p<0.01$). It implies that appearance of the product among treatment varies in favor of Treatment C. It can be said the Treatment C was highly attractive compared to other Treatments.

When aroma was considered in the grouping, the same Table shows a highly significant difference existed among Treatments ($F=4.708$; $p<0.01$). It implies that product in Treatment C was considered very much pleasant than its Treatments counterparts. It can be said that the 50-50 combination of rice flour and sweet potato flour contributed to the acceptable aroma of the product.

In terms of taste, it can be noted a highly significant differences among Treatment ($F=7.221$; $p<0.01$). It implies that the 50-50 formulation of rice and sweet potato flour were like very much by the evaluators. It can be said that the sweetness of the sweet potato added to the aftertaste of the product.

For the texture of the product, it was indicated by the F-value of 6.050 and a significant value of 0.001 that there was a highly significant difference among Treatments. It implies that the softness of the product considering its treatments varies. It can be said that Treatment C was evaluated among others as moderately soft. This result was in congruent to the result of the study of Chon and Li (2018) who revealed that significant difference were found using the proportion of sweet potato in making steamed rice bowl cake. Their findings indicated that the proportion of sweet potatoes was positively related to cohesiveness and chewiness of steamed rice bowl cake.

In terms of general acceptability of the products, data show a highly significant differences existed among Treatments ($F=7.240$; $p<0.01$). It implies that Treatment C was the product rated as highly acceptable by the evaluators.

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(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

Conclusions and Recommendations

The incorporation of sweet potato flour in the formulation of rice puto was found to improve the appearance of the product since sweet potato flour blends well with the rice flour. The pleasant aroma, sweetness, and softness property of sweet potato added a more acceptable rice-sweet potato puto.

The 50-50 formulation of sweet potato flour and rice flour was the most accepted proportion. This proportion was the perfect combination since it neutrally balances the sweet properties present in the sweet potato flour to rice flour.

The appearance, aroma, taste, texture, and general acceptability of the product among Treatment varies in favor of Treatment C.

It was recommended to utilized 50% sweet potato flour and 50% rice flour in making puto to improve its sensory qualities. The accepted Treatment should be evaluated by 100 consumer to validate its acceptability level. The product should be tested for its self life and microbial analysis.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

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(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

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(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

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(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indention with 12-point space in between.)

Arendt, E. & Zannini, E. (2013). Rice cakes. *Cereal Grains for the Food and Beverage Industries*. Sciencedirect.com

Chon, C & Li, M. (2018). A research of effect of three sweet potato varieties and addition on resistant starch content and physical characteristics of steamed rice bowl cake. *Journal of Food and Nutrition Research*, 6 (9), 551-556.DOI:10.1269

Saeed, S., Ahmad, M., Kausar, H., Parveen, S., Masih, S., & Salam, A. (2012). Effect of sweet potato flour on quality of cookies. *Food Technology Section, Post Harvest Research Centre, Ayub Agricultural Research Institute*. DOI:10.13140/RG.2.2.28963.302

Shih, F. Truong, V. & Daigle, K. (2005). Physicochemical properties of gluten-free pancakes from rice and sweet potato flours. *Journal of Food Quality*, 29 (1), 1-107

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Wijaya, H., Slay. A. & Abdullah, N. (2021). Ice cream products made from processed purple sweet potatoes: a product organoleptic study. *IOP Conf. Series: Earth and Environmental Science*, 807 042074IOP doi:10.1088/1755-1315/807/4/04207